Statistics Qualifying Examination

Answer all questions and show all work.
This exam is closed-note/book. You need to use a calculator.

1. Suppose that Y_1 and Y_2 have the joint distribution

$$f_{Y_1,Y_2}(y_1, y_2) = \begin{cases} \frac{y_1 y_2}{2} & \text{if } 0 \leq y_2 \leq y_1 \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

   (a) Find the conditional probability density function of Y_2 given $Y_1 = y_1$.

   (b) Find the probability density function of $U = Y_1 - Y_2$.

2. Suppose that a random variable Z follows the standard normal distribution.

   (a) Prove that the moment generating function of Z is $M_Z(t) = e^{t^2/2}$.

   (b) Suppose that Y is a random variable such that $Y = \mu + \sigma Z$ where μ and σ are unknown, but fixed constants.

   Using the moment generating function of Z in part (a), show that the moment generating function of Y is $M_Y(t) = e^{\mu t + \sigma^2 t^2/2}$.

3. Let X have a Poisson distribution with $\mu = 100$. Use Chebyshev's inequality to determine a **lower bound** for $P(75 < X < 125)$.

4. Let $X_n \sim N\left(\mu, \sigma + \frac{1}{n}\right)$, where μ and σ are both finite, and $\sigma > 0$.

   (a) Determine whether X_n converges in distribution as $n \to \infty$. If so, identify the limiting random variable and its cumulative distribution function, and justify your answer carefully.

   (b) Determine whether X_n converges in probability to the random variable in (a) as $n \to \infty$. Clearly justify your answer.

5. Let $X_1, \ldots, X_n$ be independent and identically distributed as

$$X_i \sim \mathrm{Exp}(1/\theta), \qquad \theta > 0,$$

   so that the density is

$$f(x;\theta) = \theta e^{-\theta x}, \qquad x > 0.$$

   (a) Derive the Cramér–Rao lower bound for any unbiased estimator of θ.

   (b) Find the minimum variance unbiased estimator (MVUE) of θ. Determine whether the MVUE is efficient, and justify your answer.

   (c) Derive the level-α likelihood ratio test for

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_1 : \theta \neq \theta_0,$$

   and express the rejection region in terms of $T = \sum_{i=1}^{n} X_i$.

6. Chateau Latour is widely acknowledged as one of the world's greatest wine estates with a rich history dating back to at least 1638. The Grand Vin de Chateau Latour is a wine of incredible power and longevity. At a tasting in New York in April 2000, the 1863 and 1899 vintages of Latour were rated alongside the 1945 and the 1961 vintages as the best in a line-up of 39 vintages ranging from 1863 to 1999 (Wine Spectator , August 31, 2000). Quality of a particular vintage of Chateau Latour has a huge impact on price. For example, in March 2007, the 1997 vintage of Chateau Latour could be purchased for as little as $159 per bottle while the 2000 vintage of Chateau Latour costs as least $700 per bottle (www.wine-searcher.com).

   Harvest began on the 13th September and lasted on the 29th, frequently interrupted by storm showers. But quite amazingly the dilution effect in the grapes was very limited ?. (http://www.chateau-latour.com/commentaires/1994uk.html? ; Accessed: March 16, 2007) Each vintage was classified as having had "unwanted rain at harvest" (e.g., the 1994 vintage) or not (e.g., the 1996 vintage) on the basis of information like that reproduced above. Thus, the data consist of:

   - Vintage = year the grapes were harvested
   - Quality – on a scale from 1 (worst) to 5 (best) with some half points
   - End of harvest – measured as the number days since August 31
   - Rain – a dummy variable for unwanted rain at harvest = 1 if yes.

   The first model considered was:

$$\text{Quality} = \beta_0 + \beta_1 \times \text{End of Harvest} + \beta_2 \times \text{Rain} + \beta_3 \times \text{Rain} \times \text{End of Harvest} + \epsilon$$

   Use t-values: $t_{0.01,40} = 2.4233$, $t_{0.025,40} = 2.021$, $t_{0.05,40} = 1.684$, $t_{0.10,40} = 1.303$, $t_{0.025,5} = 2.7764$, $t_{0.05,5} = 2.1318$.

```
Call:
lm(formula = Quality ~ EndofHarvest + Rain + Rain:EndofHarvest)

Residuals:
Min      1Q  Median      3Q     Max
-1.6833 -0.5703  0.1265  0.4385  1.6354

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         5.16122    0.68917   7.489 3.95e-09 ***
EndofHarvest       -0.03145    0.01760  -1.787   0.0816 .
Rain                1.78670    1.31740   1.356   0.1826
EndofHarvest:Rain -0.08314    0.03160  -2.631   0.0120 *
```

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Residual standard error: 0.7578 on 40 degrees of freedom
Multiple R-squared:  0.6848,    Adjusted R-squared:  0.6612
F-statistic: 28.97 on 3 and 40 DF,  p-value: 4.017e-10
```

(a) Write down the expressions for the fitted regression lines for the two cases: when there is "unwanted rain at harvest" and when there is **no** "unwanted rain at harvest". Please use numerical estimates (not symbolic coefficients such as $\hat{\beta}$).

(b) Interpret the estimates of $\hat{\beta}_0$ and $\hat{\beta}_3$. Provide your interpretations in words only (do not use Y or X notation).

(c) Assume that the modeling assumptions hold. After fitting the regression model, the researchers wish to test the null hypothesis $H_0 : \beta_2 \leq 1.2$ against the alternative $H_A : \beta_2 > 1.2$. Conduct the appropriate hypothesis test at significance level $\alpha = 0.05$, clearly stating the test statistic, decision rule, and conclusion.

(d) Estimate the number of days of delay to the end of harvest it takes to decrease the quality rating by 1 point when there is:

- (i) No unwanted rain at harvest
- (ii) Some unwanted rain at harvest

7. Some researchers observed the data on 20 individuals with high blood pressure. The researchers were interested in determining if a relationship exists between blood pressure and age, weight, body surface area, duration, pulse rate and/or stress level.
The following variables are recorded:

- blood pressure (Y = BP, in mm Hg)
- age ($X1$ = Age, in years)
- weight ($X2$ = Weight, in kg)
- body surface area ($X3$ = BSA, in sq m)
- duration of hypertension ($X4$ = Dur, in years)
- basal pulse ($X5$ = Pulse, in beats per minute)
- stress index ($X6$ = Stress)

The summary of the multiple regression analysis is given below:

```
Call:
lm(formula = Y ~ X1 + X2 + X3 + X4 + X5 + X6)

Residuals:
Min       1Q   Median       3Q      Max
-0.93213 -0.11314  0.03064  0.21834  0.48454

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.870476   2.556650  -5.034 0.000229 ***
X1            0.703259   0.049606  14.177 2.76e-09 ***
X2            0.969920   0.063108  15.369 1.02e-09 ***
X3            ********   ********   ***** ******** *
X4            0.068383   0.048441   1.412 0.181534
X5           -0.084485   0.051609  -1.637 0.125594
X6            0.005572   0.003412   1.633 0.126491
---

Residual standard error: 0.4072 on 13 degrees of freedom
Multiple R-squared:  0.9962,    Adjusted R-squared:  0.9944
F-statistic: 560.6 on 6 and 13 DF,  p-value: 6.395e-15
```

The researchers are also given the F-tables below as well as the best subset regression using the R function *regsubsets* with *nbest=4* to perform best-subset based model selection and *summaryHH* (in the next page). Output is summarized below. *rss* denotes the sum of squares of errors/residuals (SSE). Also *rsq* denotes the R^2.

(df1, df2)	(1,13)	(1,14)	(1,15)	(1,16)	(1,17)	(1,18)	(1,19)
$F_{0.025;df1,df2}$	6.414	6.298	6.199	6.115	6.042	5.978	5.922
$F_{0.05;df1,df2}$	4.667	4.600	4.543	4.493	4.451	4.414	4.381

(df1, df2)	(2,13)	(2,14)	(2,15)	(2,16)	(2,17)	(2,18)	(2,19)
$F_{0.025;df1,df2}$	4.965266	4.856698	4.765048	4.686665	4.618874	4.559672	4.507528
$F_{0.05;df1,df2}$	3.805565	3.738892	3.682320	3.633723	3.591531	3.554557	3.521893

```
> library(leaps)
> BP.bestreg<-regsubsets(Y~., data=data1, method="exhaustive",nbest=4)
> library(HH)
> # S3 method for regsubsets
> summaryHH(BP.bestreg)
               model p   rsq    rss adjr2      cp    bic stderr
1                 X2 2 0.903  54.53 0.897  312.81 -40.59  1.740
2                 X3 2 0.750 140.14 0.736  829.07 -21.71  2.790
3                 X5 2 0.520 268.56 0.494 1603.41  -8.71  3.863
4                 X1 2 0.434 316.73 0.403 1893.93  -5.41  4.195
5              X1-X2 3 0.991   4.82 0.990   15.09 -86.10  0.533
6              X2-X6 3 0.920  44.87 0.910  256.56 -41.50  1.625
7              X2-X5 3 0.919  45.59 0.909  260.90 -41.18  1.638
8              X2-X4 3 0.914  48.43 0.903  278.06 -39.97  1.688
9           X1-X2-X3 4 0.995   3.06 0.994    6.43 -92.23  0.437
10          X1-X2-X5 4 0.992   4.33 0.991   14.10 -85.27  0.520
11          X1-X2-X4 4 0.992   4.65 0.990   16.01 -83.86  0.539
12          X1-X2-X6 4 0.992   4.66 0.990   16.11 -83.79  0.540
13       X1-X2-X3-X4 5 0.995   2.72 0.994    6.41 -91.56  0.426
14       X1-X2-X3-X6 5 0.995   2.84 0.994    7.12 -90.72  0.435
15       X1-X2-X3-X5 5 0.995   3.02 0.993    8.18 -89.51  0.448
16       X1-X2-X5-X6 5 0.994   3.39 0.992   10.47 -87.14  0.476
17    X1-X2-X3-X5-X6 6 0.996   2.49 0.994    6.99 -90.37  0.421
18    X1-X2-X3-X4-X5 6 0.995   2.60 0.994    7.67 -89.49  0.431
19    X1-X2-X3-X4-X6 6 0.995   2.60 0.994    7.68 -89.47  0.431
20    X1-X2-X4-X5-X6 6 0.994   3.10 0.992   10.71 -85.94  0.471
21 X1-X2-X3-X4-X5-X6 7 0.996   2.16 0.994    7.00 -90.22  0.407
```

(a) Is it worth including $X5$ in the model that already contains $X1$, $X2$, $X3$, $X4$, and $X6$? To get full credit, give the hypotheses, test statistic, degrees of freedom, (range of) p-values, and your conclusion. Use $\alpha = 0.05$.

(b) Is it worth including $X5$ & $X6$ in the model that already contains $X1$, $X2$ and $X3$? To get full credit, give the hypotheses, test statistic, degrees of freedom, (range of) p-values, and your conclusion. Use $\alpha = 0.05$.

(c) Is it worth including $X1$ in the model that does not contain any variables (null model)? To get full credit, give the hypotheses, test statistic, degrees of freedom, (range of) p-values, and your conclusion. Use $\alpha = 0.05$.

(d) In the full model, the one containing all the independent variables ($X1$, $X2$, $X3$, $X4$, $X5$, and $X6$), the least square error estimators of $\hat{\beta}_3$ is positive. Given this information as well as the summary tables of the R output above, calculate the value of $\hat{\beta}_3$.

8. Disk drive substrates may affect the amplitude of the signal obtained during readback. A manufacturer compares four substrates: aluminum(A), nickel-plated aluminum (B), and two types of glass (C and D). Sixteen disk drives will be made, four using each of the substrates. The design responses (in microvolts) are given in the following table (data from Nelson 1993; Greek letters indicate day)

	Operator			
Machine	1	2	3	4
1	$A\alpha = 8$	$C\gamma = 11$	$D\delta = 2$	$B\beta = 8$
2	$C\delta = 7$	$A\beta = 5$	$B\alpha = 2$	$D\gamma = 4$
3	$D\beta = 3$	$B\delta = 9$	$A\gamma = 7$	$C\alpha = 9$
4	$B\gamma = 4$	$D\alpha = 5$	$C\beta = 9$	$A\delta = 3$

The grand mean is $\bar{Y}_{...} = 6$, and the level means for the four substrates are:

$$A : 5.75 \quad B : 5.50 \quad C : 9.00 \quad D : 3.75$$

(a) What kind of design is used for the experiment? Give a model and state the assumptions.

(b) Part of the ANOVA table from SAS is given below. Test if the *substrates* are different from each other in terms of their effects on the response. To get full credits, state the hypotheses, obtain the test statistic, and draw your conclusion ($\alpha = 0.05$).

Source	DF	Squares	Mean Square	F Value	Pr > F
Model	12	100.5000000	8.3750000	1.17	0.5098
Error	3	21.5000000	7.1666667		
Corrected Total	15	122.0000000			

Source	DF	Type I SS	Mean Square	F Value	Pr > F
row	??	21.50000000	7.16666667	1.00	0.5000
col	??	14.00000000	4.66666667	0.65	0.6335
substrate	??	**??**	**??**	**??**	**??**
greek	??	3.50000000	1.16666667	0.16	0.9149

(df_1, df_2)	$(3,3)$	$(3,12)$	(3,15)	$(4,3)$	$(4,12)$	(4,15)
$F_{0.025;df_1,df_2}$	15.4392	4.4742	4.1528	15.1010	4.1212	3.8043
$F_{0.05;df_1,df_2}$	9.2766	3.4903	3.2874	9.1172	3.2592	3.0556

(c) If the machine were not considered as blocks and not included in ANOVA model, will the test results in part (c) change? Justify your answer.

9. An engineer suspects that the surface finish of a metal part is influenced by the feed rate and the depth of cut. He selects three feed rates and four depths of cut. He then conducts a factorial experiment and obtains the following data.

	Depth of Cut (in)			
Feed rate (in/min)	1	2	3	4
1	74, 64, 60	79, 68, 73	82, 88, 92	99, 104, 96
2	92, 86, 88	98, 104, 88	99, 108, 95	104, 110, 99
3	99, 98, 102	104, 99, 95	108, 110, 99	114, 111, 107

Some summary statistics are give on the next page.

```
Grand mean: 94.333

feed        mean     |    depth   mean
1          81.583    |    1       84.778
2          97.583    |    2       89.778
3         103.833    |    3       97.889
                     |    4      104.889
------------------------------------------------
feed    depth       mean
1       1          66.000
1       2          73.333
1       3          87.333
1       4          99.667
2       1          88.667
2       2          96.667
2       3         100.667
2       4         104.333
3       1          99.667
3       2          99.333
3       3         105.667
3       4         110.667
```

Suppose the following statistical model is used to fit the data.

$$Y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \epsilon_{ijk}; k = 1, 2, 3$$

where τ_i $(i = 1, 2, 3)$ and β_j $(j = 1, 2, 3, 4)$ are the effects of feed rate and cut depth, and $(\tau\beta)_{ij}$ are their interactions. For parameter estimation, we impose the following constraints as in the lecture notes: $\sum_i \tau_i = \sum_j \beta_j = \sum_i (\tau\beta)_{ij} = \sum_j (\tau\beta)_{ij} = 0$.

The ANOVA of the data was done in SAS and the output is shown on the next page.

```
Dependent Variable: finish
                  Sum of
Source     DF    Squares       Mean Square    F Value  Pr > F
Model      11     5842.667     531.151515      18.49    <.0001
Error      24      689.333      28.722222
Co Total   35     6532.000

ource      DF     Type I SS     Mean Square F Value    Pr > F
feed        2     3160.500      1580.250      55.02    <.0001
depth       3     2125.111       708.370      24.66    <.0001
feed*depth 6       557.056        92.843       3.23    0.0180
```

(a) What are the estimates of τ_1 and $(\tau\beta)_{22}$?

(b) Test if the interaction between feed rate and cut depth is significant. To get full credits, give a test statistic, the corresponding p-value, and your conclusion.

(c) If we plan to perform pairwise comparison for *all* treatment combinations, which procedure should we use? What is the corresponding critical difference (using $\alpha = 5\%$)? *Note:* No table of critical values from distributions is given for this part, and thus you don't need to calculate the final numerical answer. Instead, please give the formula of the critical difference and specify the components in the formula, including but not limited to the degrees of freedom, significance level, and the distribution.

(d) Use the Bonferroni method to compare the following treatments (i.e. these *four specific* level combinations of speed and depth): (2,3), (2,4), (3,3) and (3,4), *pairwisely* (Use $\alpha = 6\%$). Calculate the critical difference and report your results of comparison. You can report the result as we have seen in SAS output by labeling significantly different combinations with different Latin letters.

df	2	3	6	24
$t_{0.005;df}$	9.9248	5.8409	3.7074	2.7969
$t_{0.01;df}$	6.9646	4.5407	3.1427	2.4922
$t_{0.015;df}$	5.6428	3.8960	2.8289	2.3069
$t_{0.03;df}$	3.8964	2.9505	2.3133	1.9740